

GEOTeams: Piloting an innovative multilevel team approach to creating summer research pathways to geoscience

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Abstract. ABSTRACT The geoscience profession must attract and retain more geoscience students in secondary and postsecondary educational settings to address a projected shortage of geoscience professionals in the United States in the next decade. The geoscience workforce is also among the least diverse of all the science, technology, engineering, and mathematics (STEM) workforce populations in the United States, motivating efforts to increase participation by women and nonbinary people, Hispanics or Latinos, Blacks or African Americans, and American Indians or Alaska Natives. National Science Foundation award 1911514 established the “GEOTeams” program, which tested an approach to broadening participation in geoscience education through intensive team-based summer research experiences, skills workshops, and career networking seminars. Over three summers, a total of 36 participants worked in four teams per year, each composed of a high school student, an undergraduate STEM major, and a preservice STEM teacher, undertaking geoscience research under the guidance of a faculty member in the Hofstra University Department of Geology, Environment, and Sustainability. Evaluation of the mixed-level team-based cohort model used in the GEOTeams program showed that this approach provides engaging and effective introductions to geoscience research and careers for all participants. The project also met its goal of including a significant number of female and underrepresented minority participants in each annual cohort. We hope that our successful pilot of this model will stimulate further research on the value of teamwork in geoscience training and encourage institutions to expand participation in undergraduate summer research experiences to include high school students and preservice teachers.